

1. Data Mining is the process of extracting useful patterns, trends and knowledge from large volumes of data using statistical and machine learning techniques.

Two basic data mining tasks:

Classification → Assigns data into predefined categories

Clustering → Groups similar data objects without predefined labels.

2. KDD (Knowledge Discovery in Databases) is the process of discovering useful knowledge from data.

Steps of KDD:

- Data cleaning
- Data integration
- Data selection
- Data Mining
- Pattern Evaluation

Importance of Data preprocessing

- Removes errors and missing values.
- Improves the accuracy and efficiency of data mining.

3. Data Cleaning is the process of detecting and correcting inaccurate, incomplete or inconsistent data.

Two data quality issues:

- Missing values
- Noisy (incorrect) data

4. Missing values:

- Replace with mean, median or mode.
- Predict values using machine learning methods.

Noisy data:

- Binning
- Regression
- Clustering

These methods improve data quality before mining.

5. Data Integration is the process of combining data from multiple sources into a single, consistent database.

Challenges:

- Data redundancy (duplicate data)
- Data inconsistency (different formats or values)

6. Covariance measures the relationship b/w variables and indicates whether they move together.

population Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

Sample Covariance:

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

7. A typical Data mining system consists of:

- Data base / Data Warehouse
- Data cleaning and integration
- Data mining engine
- pattern Evaluation module.

It converts raw data into useful knowledge.

8. The major steps are:

- Data cleaning
- Data integration
- Data Transformation
- Data Reduction

These steps improve data quality and reduce processing time

9. Equal - Frequency Binning is a discretization method in which each bin contains an equal number of data values.

Advantages:

- Reduces noise
- Handles skewed data effectively
- Improves data analysis.

10. Data Discretization is the process of converting continuous data into discrete intervals (bins)

Equal-Width Binning

equal interval size

Simple to implement

Sensitive to outliers

Equal-Frequency Binning

Equal number of values

Better for skewed data

Less affected by outliers